Fabrics

Fabrics are materials made from fibers obtained naturally or via chemical processing. A wide range of fabrics are manufactured, each with varying properties that may suit different needs, such as crease resistance, durability, water resistance, and elasticity.

Raw materials

Fibers for fabrics come from many natural sources, including plants grown as crops, such as cotton and flax, and animals such as sheep. Fossil fuel industries produce polymers (see p.78) that make a wide range of synthetic fabrics, including acrylic and polyester. Many of these are forced through a device called a spinneret to create long filaments that can be processed into yarn. The yarn is then knitted, woven, or bonded (see p.129).

WHAT IS THE MOST COMMON FABRIC IN THE WORLD?

Cotton makes up 30 percent of all the fibers produced for fabric. Cotton-growing accounts for 2.5 percent of global farmland use.

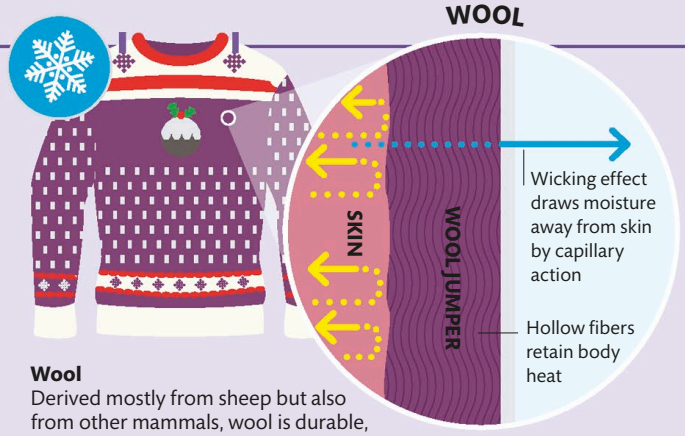
ANIMAL FIBERS



Leather
The tanned hides of animals make a material that is tough, durable, and does not tear easily. It is wind- and water-resistant but hard to sew.



Silk
Made from threads spun by silkworms, silk is lightweight, strong, offers good insulation, and holds its shape well.

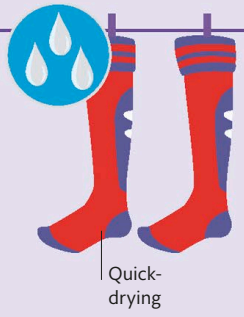


Wool
Derived mostly from sheep but also from other mammals, wool is durable, moisture-repelling, and resistant to wrinkles and soiling. It also retains heat well and draws sweat away from the skin.

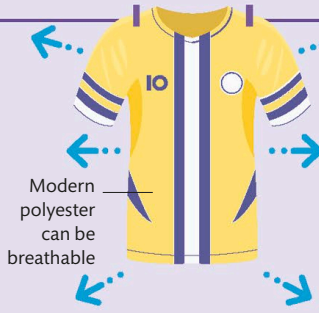
SYNTHETICS



Nylon
A synthetic material derived from coal, nylon is made into smooth, lightweight, and highly elastic fabrics.



Acrylic
While lacking a natural feel, acrylic fabrics offer good insulation, wash easily, and retain their shape well.



Polyester
Produced from petroleum, this is resistant to stretching and shrinking and barely absorbs moisture.

HEATING ELEMENTS IN SOME COATS HELP THE WEARER TO KEEP WARM

